

SMART BLIND STICK

Obstacle · Voice · Puddle Alert - Assistive Technology for the Visually Impaired

Science Exhibition | Theme: Future Technology & Robotics | Arduino UNO R3

The Problem

A normal walking stick only finds an obstacle **after** the user bumps into it. Visually impaired people need to know about obstacles - and wet, slippery floors - **before** contact, to walk safely.

Hypothesis & Aim

Hypothesis: A sensor-based stick can detect nearby obstacles before contact and warn the user through sound, voice and vibration.

Aim: Build a low-cost smart stick that helps visually impaired people walk more safely and independently.

Materials Used

- Arduino UNO R3 (the controller)
- HC-SR04 ultrasonic sensor
- Passive buzzer (rising-pitch beep)
- Vibration motor (handle buzz)
- ISD1820 voice module + speaker
- Rain/water sensor (puddle alert)
- Warning LED + 220 Ω resistor
- USB power bank · walking stick / PVC pipe

How It Works

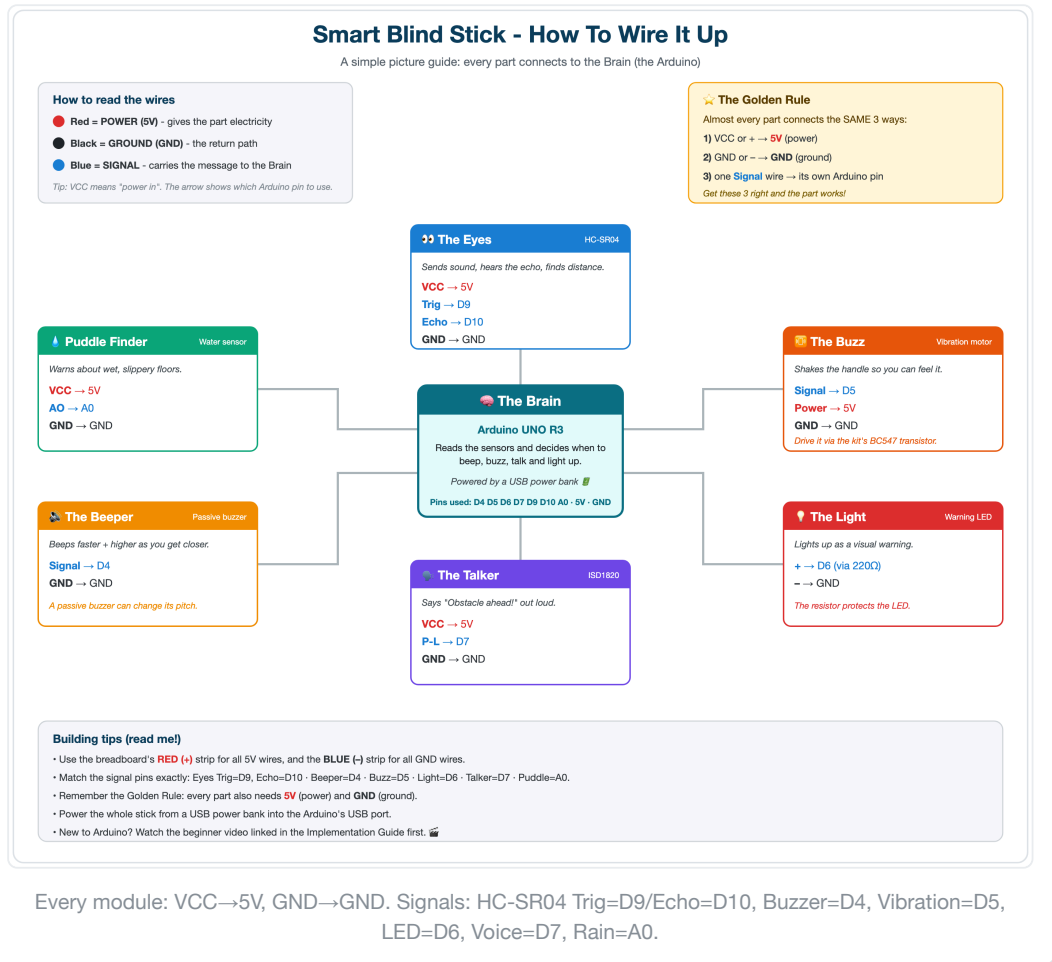
The ultrasonic sensor works like a **bat's sonar**. It sends out sound waves; when they hit an object they bounce back. The Arduino measures the echo time and calculates the distance.

$$\text{Distance} = (\text{Speed of sound} \times \text{Time}) \div 2$$

$\div 2$ because the sound travels to the obstacle AND back

The closer the obstacle, the **faster and higher** the beep. Very close $\rightarrow$ a solid tone, the handle vibrates, and a voice says "*Obstacle ahead.*"

Circuit & Wiring



Alert Levels

Distance to obstacle	Alert
more than 150 cm	silent (clear)
150 $\rightarrow$ 20 cm	beeps get faster + higher
under 20 cm	solid tone + vibration + voice
wet floor / puddle	distinct fast warble

Results & Data

We tested the stick at measured distances (20 trials):

Actual	Expected	Result
100 cm	slow beep	✓
45 cm	fast beep	✓
18 cm	solid + voice	✓
wet floor	warble	✓

$$\text{Accuracy} = 18 / 20 \times 100 = 90\%$$

Misses happen on soft / angled surfaces that absorb ultrasound.

Applications

- Blind aid
- Elderly support
- Indoor navigation
- Monsoon safety
- Low-vision walking

A low-cost device (under ₹4,000) that solves a real, everyday human problem.

Future Scope

- GPS + GSM "SOS" button for emergencies
- Fall detection (motion sensor)
- Phone app with live alerts
- Camera + AI to name the obstacle

Why It Matters

It turns an ordinary stick into a **smart, sensing companion** - warning the user early, in three ways at once, so they can walk with more confidence and safety.

Smart Blind Stick - Full Wiring Diagram

High-resolution vector reference - every part connects to the Arduino.

Smart Blind Stick - How To Wire It Up

A simple picture guide: every part connects to the Brain (the Arduino)

How to read the wires

-  **Red = POWER (5V)** - gives the part electricity
-  **Black = GROUND (GND)** - the return path
-  **Blue = SIGNAL** - carries the message to the Brain

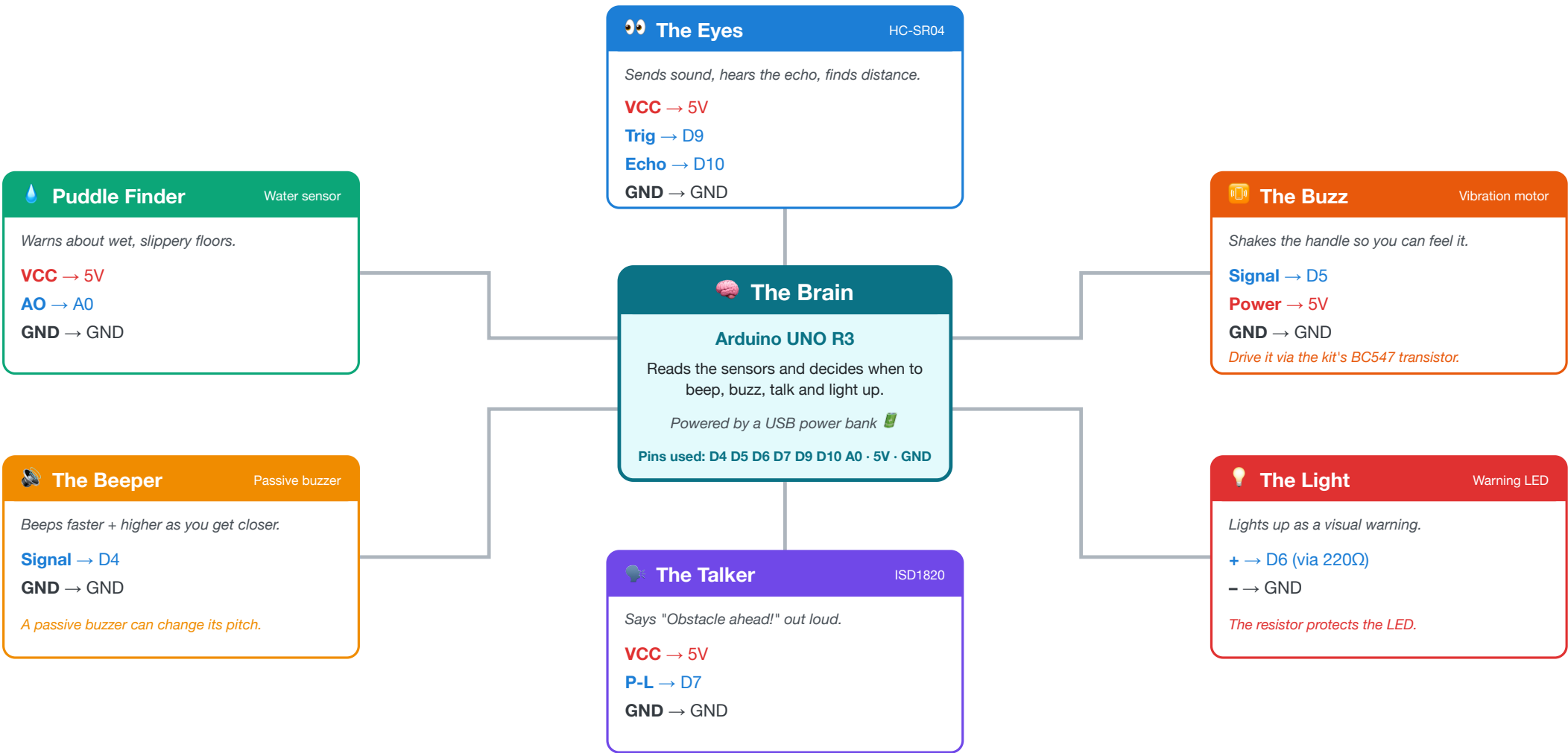
Tip: VCC means "power in". The arrow shows which Arduino pin to use.

★ The Golden Rule

Almost every part connects the SAME 3 ways:

- 1) VCC or + → **5V** (power)
- 2) GND or - → **GND** (ground)
- 3) one **Signal** wire → its own Arduino pin

Get these 3 right and the part works!



Building tips (read me!)

- Use the breadboard's **RED (+)** strip for all 5V wires, and the **BLUE (-)** strip for all GND wires.
- Match the signal pins exactly: Eyes Trig=D9, Echo=D10 · Beeper=D4 · Buzz=D5 · Light=D6 · Talker=D7 · Puddle=A0.
- Remember the Golden Rule: every part also needs **5V** (power) and **GND** (ground).
- Power the whole stick from a USB power bank into the Arduino's USB port.
- New to Arduino? Watch the beginner video linked in the Implementation Guide first. 📺